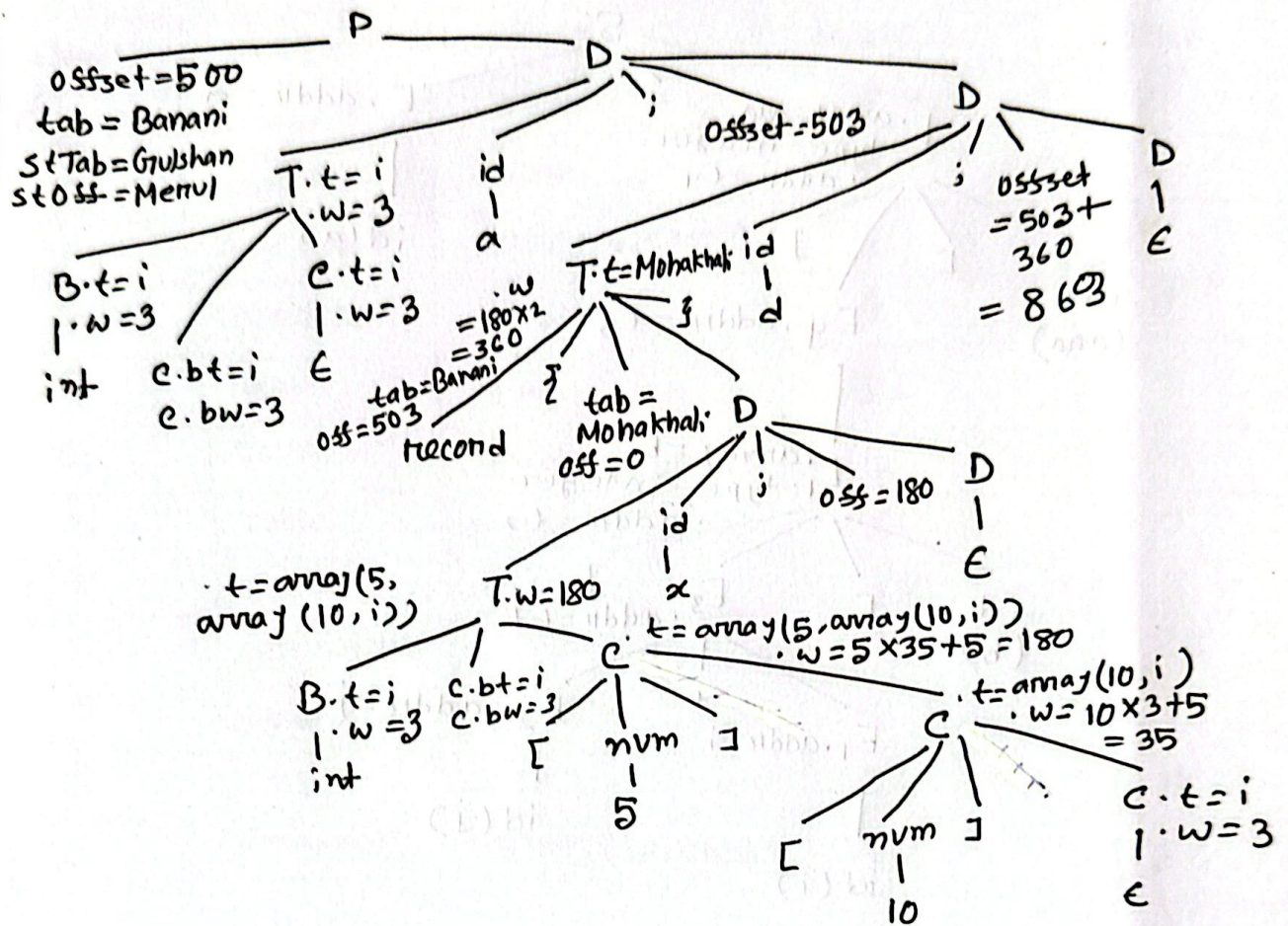


Spring 2026 (A)

7 a)



$(a, \{type: i, width: 3, offset: 503\})$
tab(Banani)

Banani
stTAB(Gulshan)

503
stOff(Merrul)

$(x, \{Type: array(5, array(10, i)), width: 180, offset: 0\})$
tab(Mohakhali)

$(d, \{type: Mohakhali, width: 360, offset: 503\})$
tab(Banani)

④

Struct	Name	Type	Width	Offset
P	Grade	char	1	0
P	marks	int	4	1
P	section	array(6, char)	6	5
Q	courseld	int	4	0
Q	result	(P)	11	4
Q	room	array(5, char)	5	15
R	title	array(8, char)	8	0
R	course	(Q)	20	8
R	semester	int	4	28
S	studentId	int	4	0
S	info	(R)	32	4
S	university	array(12, char)	12	36

offset(info) in S = 4

offset(course) in R = 8

offset(result) in Q = 4

offset(section) in P = 5

[4] $\rightarrow 4 \times 1 = 4$

Total offset = $4 + 8 + 4 + 5 + 4 = 25$ bytes

b)

t1 = @studentRecord[4]

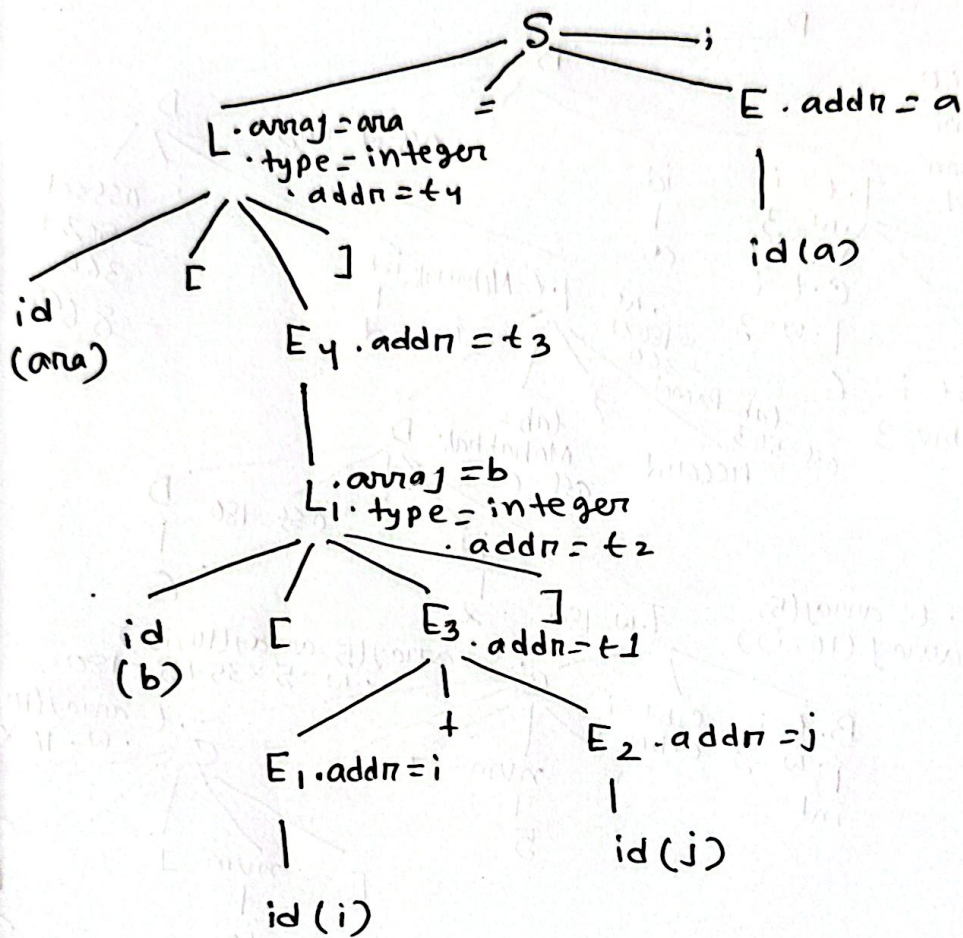
t2 = t1[8]

t3 = t2[4]

t4 = t3[5]

y = t4[4]

⑥ $ana[b[i+j]] = a;$



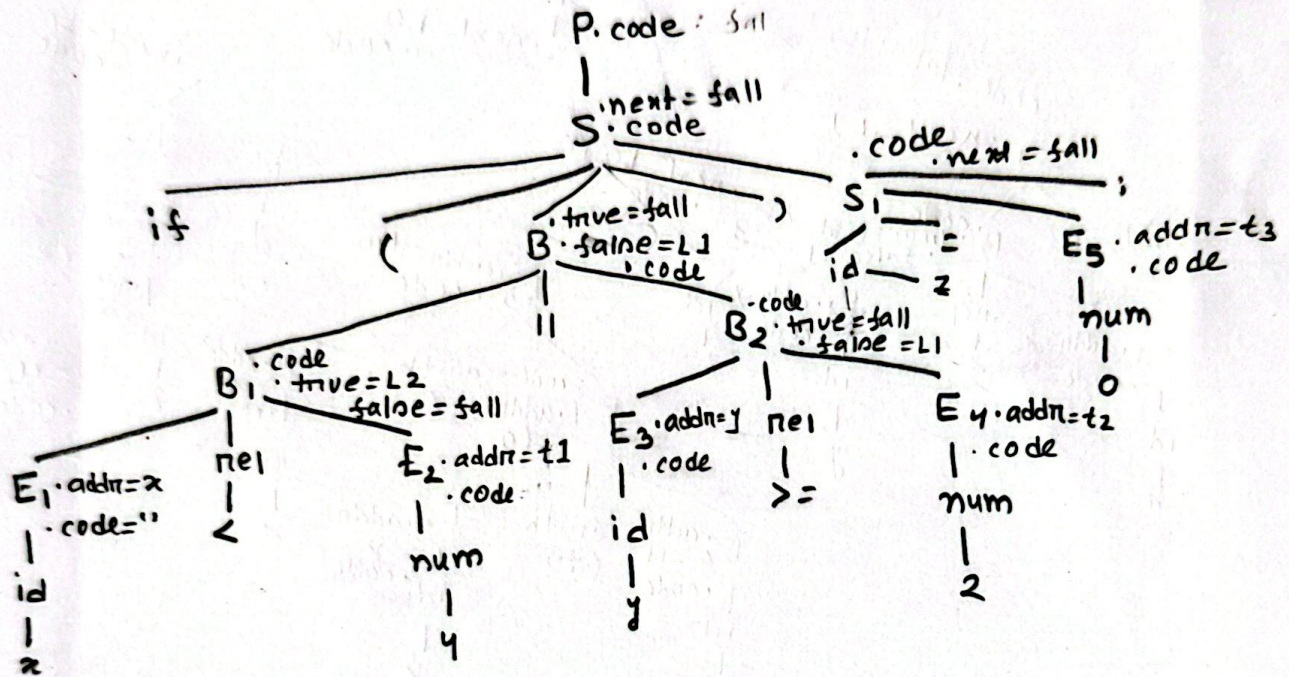
$$t_1 = i + j \quad (E_3)$$

$$t_2 = t_1 * 4 \quad (L)$$

$$t_3 = b[t_2] \quad (E_4)$$

$$t_4 = t_3 * 4 \quad (L)$$

$$ana[t_4] = a \quad (S)$$



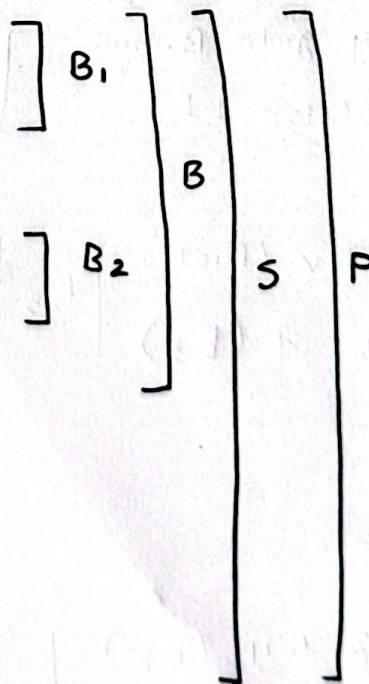
t1 = 4 (E2)
if x < t1 goto ~~B1. true~~ L2

t2 = 2 (E4)
if y not <= t2 ~~B2. false~~ L1

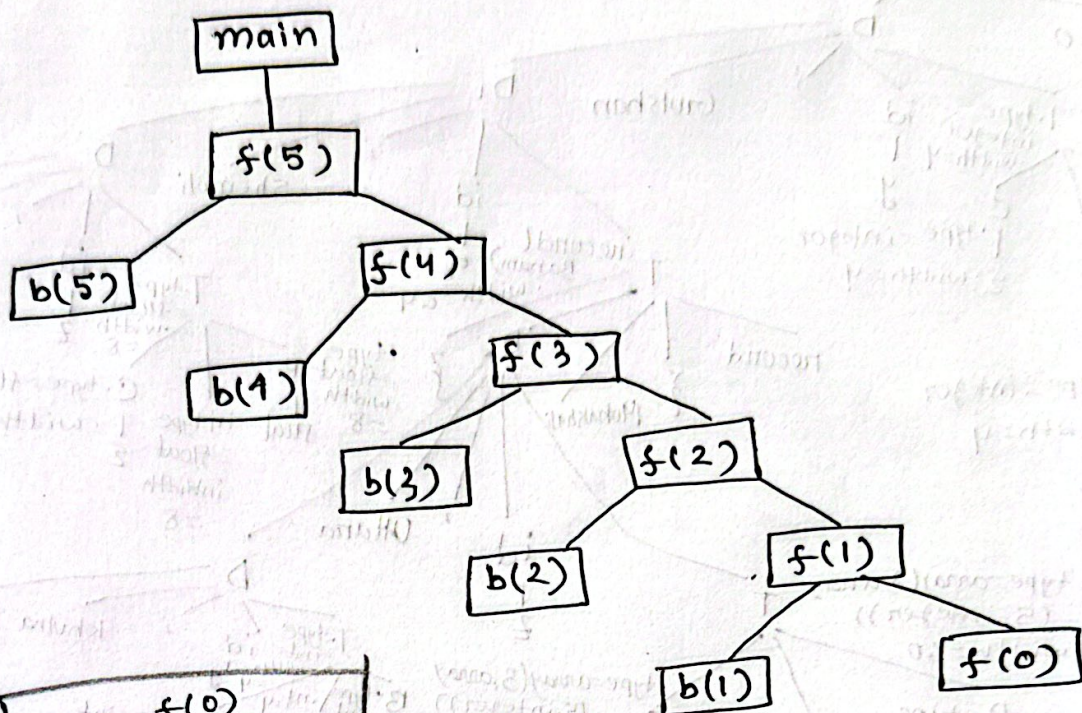
L2:

t3 = 0 (E5)
z = t3

L1:



7

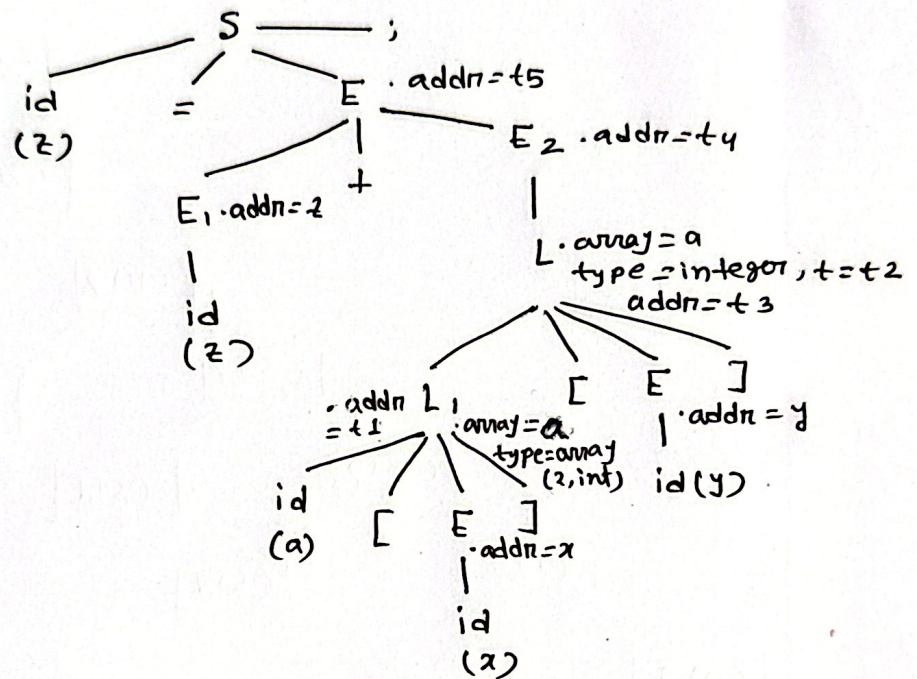


f(0)
integer n
b(1)
integer n
f(1)
integer n
b(2)
integer n
f(2)
integer n
b(3)
integer n
f(3)
integer n
b(4)
integer n
f(4)
integer n
b(5)
integer n
f(5)
integer n
integer x
main

f(3)
integer n
f(4)
integer n
f(5)
integer n
integer x
main

control stack
when execution
is inside f(3).

③ $z = z + a[x][y];$



$$t_1 = x * 8 [L_1]$$

$$t_2 = y * 4$$

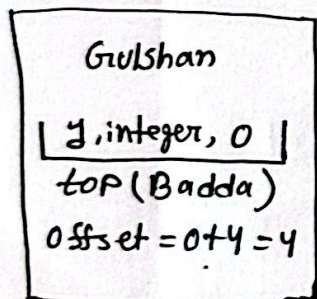
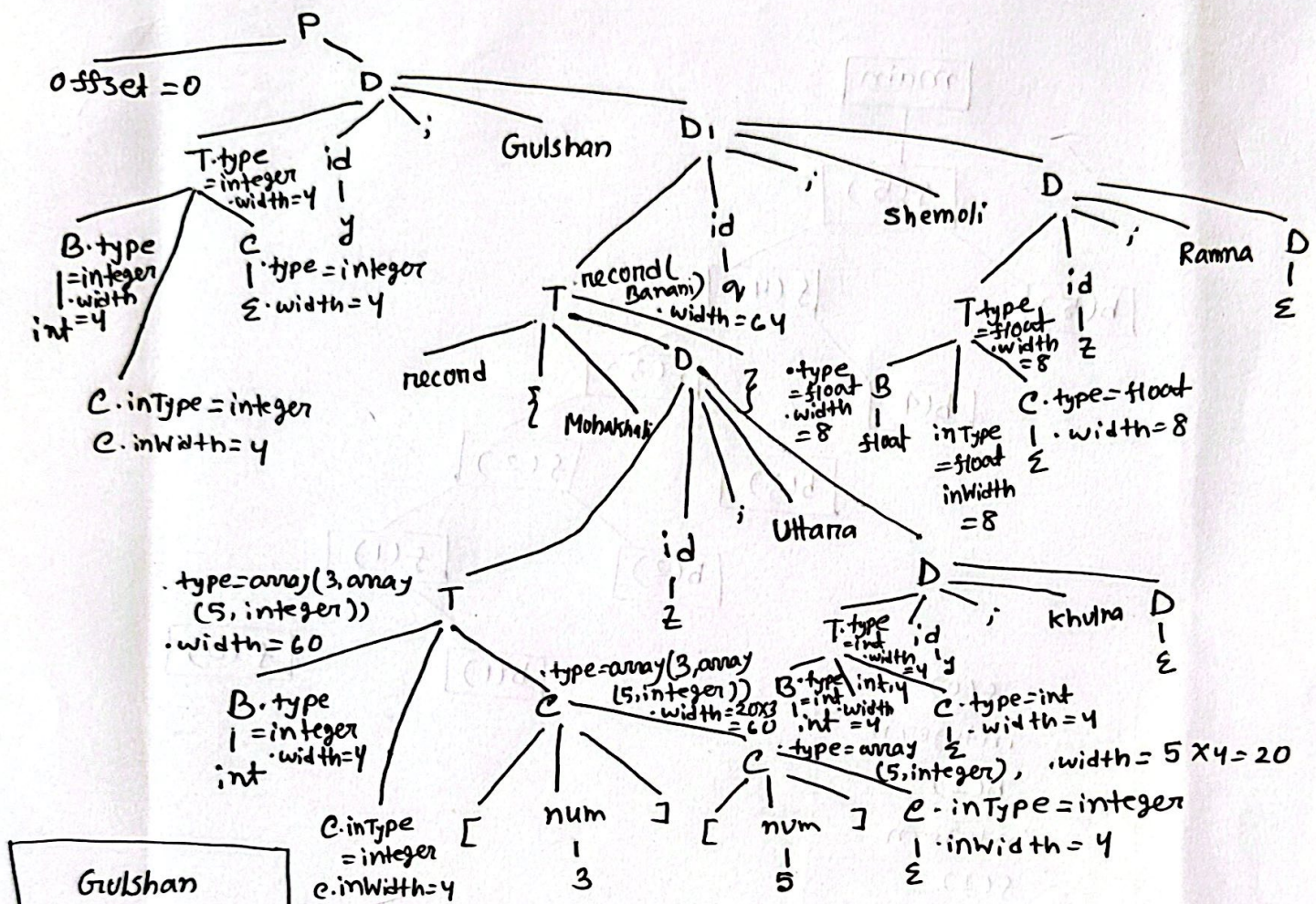
$$t_3 = t_1 + t_2$$

$$t_4 = a[t_3] [E_2]$$

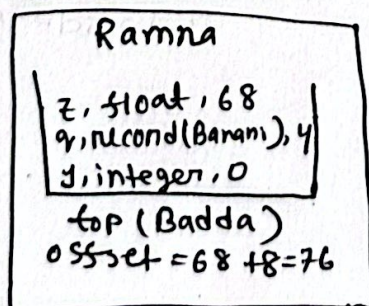
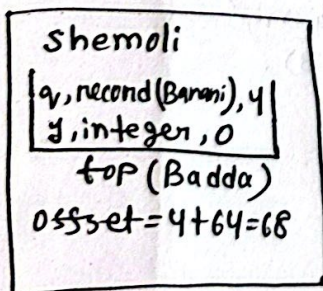
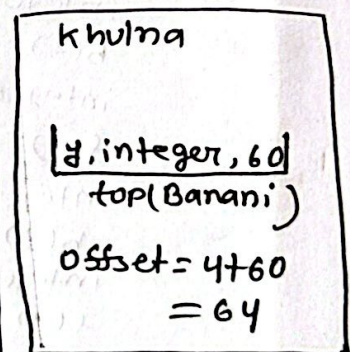
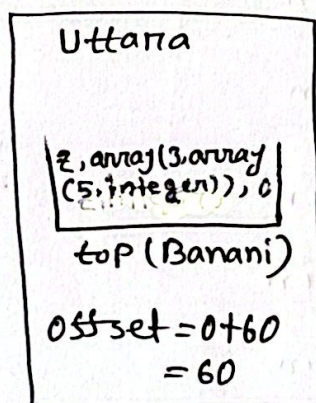
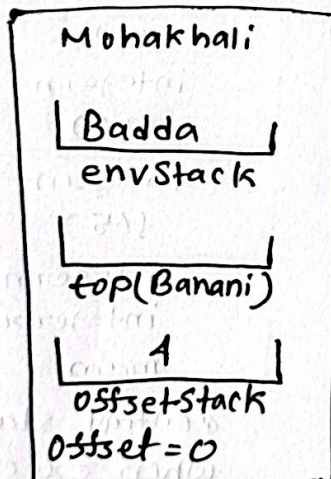
$$t_5 = z + t_4 [E]$$

$$z = t_5$$

①



```
Env rec Env = Banani  
rechWidth = 64  
top = Badda  
offset = 4
```



Spring 2025 (A)

(2)

$a = c - b * t * (-b);$

$t1 = \text{uminus } b$

$t2 = b * t$

$t3 = t2 * t1$

$t4 = c - t3$

$a = t4$

$\text{getNeuronBias}((c+d) * (-c), b * (x/t));$

$t5 = c + d$

$t6 = \text{uminus } c$

$t7 = t5 * t6$

$t8 = x / t$

$t9 = b * t8$

param $t7$

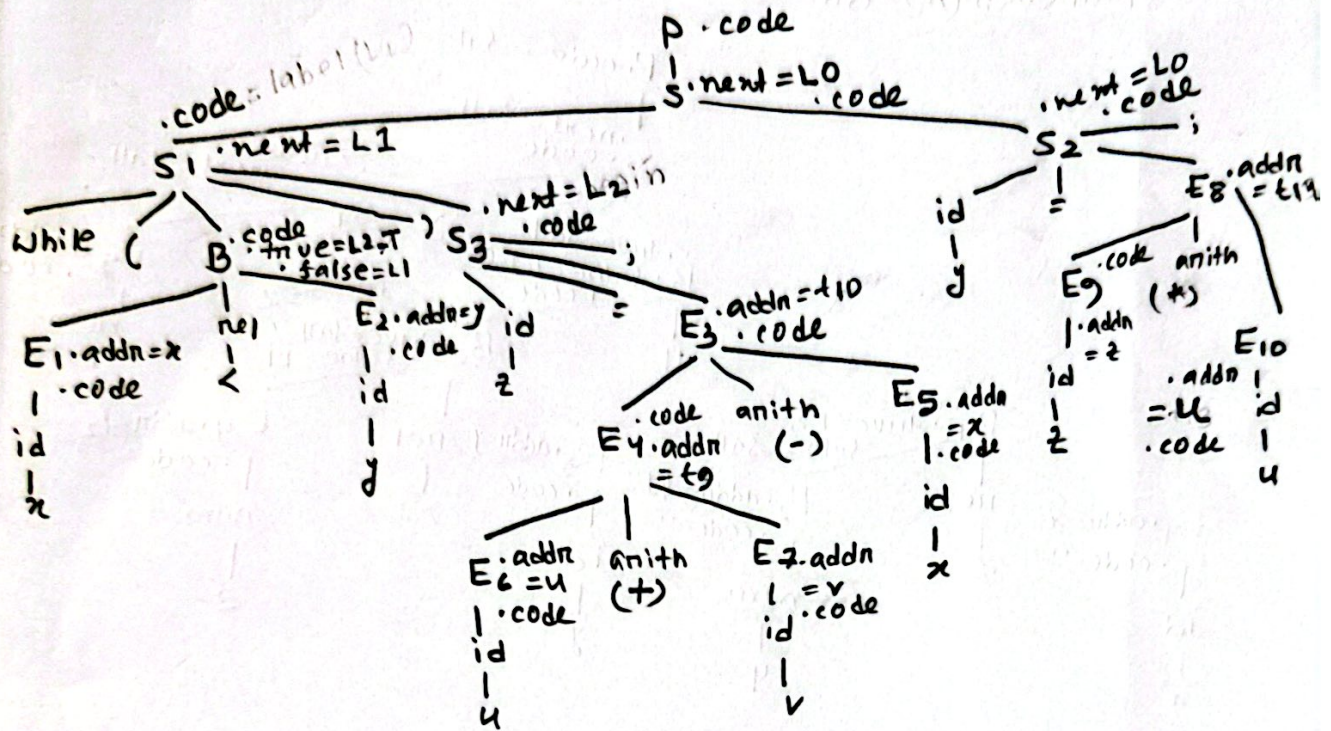
param $t9$

call $\text{getNeuronBias}, 2$

Quadruple representation

Op	arg1	arg2	result
uminus	b		t1
*	b	t	t2
*	t2	t1	t3
-	c	t3	t4
=	t4		a
+	c	d	t5
uminus	c		t6
*	t5	t6	t7
/	x	t	t8
*	b	t8	t9
param	t7		
param	t9		
call	getNeuronBias	2	

Spring 2025 (A) - Q4



begin = L2

L2:

if $x < y$ goto B ~~true~~ ^{L2-T}

goto B ~~false~~ L1

L2-T:

$t9 = u + v$ (E4)

$t10 = t9 - x$ (E3)

$z = t10$

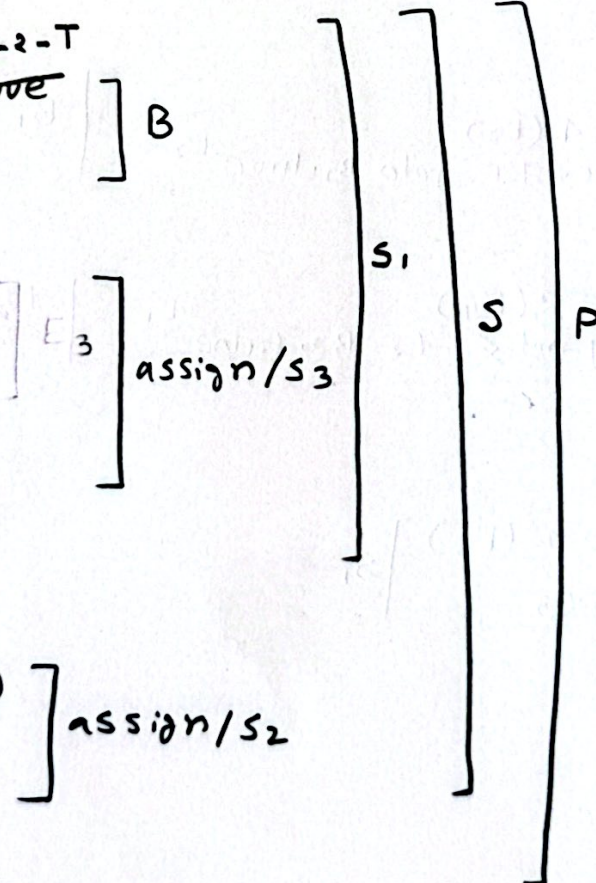
goto L2

L1:

$t14 = z * 4$ (E8)

$y = t14$

L0:



P = 0 1 2 3 4 5 6 7 8 9 10
11 12 13 14

[illegible]

Gulshan

Symbol = Gulshan(name : x)

Gulshan.type = int

Gulshan.width = 4

Gulshan.offset = 0

x, Gulshan

top(Badda)

offset = 0 + 4 = 4

Mohakhali

Badda

tabstack
(Mohakhali)

4

offsetstack

top(Barani)

offset = 0

Uttara

symbol = Uttara(name : a)

Uttara.type = int

Uttara.width = 4

Uttara.offset = 0

a, Uttara

top(Barani)

offset = 0 + 4 = 4

khulna

Barani
Badda

tabstack

4
4

offsetstack

top(Menu)

offset = 0

Shemoli

Symbol = Shemoli(name : b)

Shemoli.type = int

Shemoli.width = 4

Shemoli.offset = 0

b, Shemoli

top(Menu)

offset = 0 + 4 = 4

Ramna

Symbol = Ramna(name : c)

Ramna.type = float

Ramna.width = 8

Ramna.offset = 4

c, Ramna

b, Shemoli

top(Menu)

offset = 4 + 8 = 12

Gulistan

Symbol = Gulistan(name: w)

Gulistan.type = Menu1

Gulistan.width = 12

Gulistan.offset = 4

w, Gulistan

x, Gulistan

top(Badda)

offset = 4 + 12 = 16

Minpun

Symbol = Minpun(name: y)

Minpun.type = Menu1

Minpun.width = 16

Minpun.offset = 4

y, Minpun

a, Uthara

top(Banani)

offset = 16 + 4 = 20

Gazipur

Symbol = Gazipur(name: k)

Gazipur.type = array(5, float)

Gazipur.width = 40

Gazipur.offset = 20

k, Gazipur

y, Minpun

a, Uthara

top(Banani)

offset = 40 + 20 = 60

DMD

Symbol = DMD(name: i)

DMD.type = ~~array~~ int

DMD.width = 4

DMD.offset = 60

i, DMD

k, Gazipur

y, Minpun

a, Uthara

top(Banani)

offset = 60 + 4 = 64

③

$x = b + \text{foo}(k * 1 + 3, n);$

$t1 = k * 1$

$t2 = t1 + 3$

param $t2$

param n

$t3 = \text{call foo}, 2$

$t4 = b + t3$

$x = t4$

Quadruple :

OP	arg1	arg2	result
*	k	1	t1
+	t1	3	t2
param	t2		
param	n		
call	foo	2	t3
+	b	t3	t4
=	t4		x